# Evaluating Visual Decision Support: How Does Preference Elicitation Shape Metric Sensitivity? Supplemental Materials

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**Abstract**—Supporting decision-making is a central goal in visualization research, yet evaluating how effectively visualizations aid decision tasks remains difficult. Objective metrics for decision quality as a baseline for comparative evaluations are rare, with previous work proposing the consistency between a decision made and self-reported subjective preferences. However, the impact of preference elicitation design on such metrics is not well understood. Our focus is therefore on understanding how elicitation methods shape the sensitivity of decision quality metrics, not on comparing visualization techniques themselves. We report a preregistered study with 548 participants examining how varying the expressiveness of preference elicitation affects the sensitivity of decision accuracy metrics when comparing parallel coordinates and tabular visualizations. Our baseline condition replicates a previous study using simple preference elicitation with significantly more participants and confirms its inconclusive findings that suggest a comparable performance of parallel coordinates and tabular visualization for the tested decision task. We further test two additional conditions to investigate if more expressive preference elicitation increases the metric's sensitivity. Our results suggest the existence of a 'plateau' effect: while more expressive elicitation may improve sensitivity in detecting performance differences, further increases offer diminishing returns. We observed these sensitivity differences even though the experimental setup, with its comparable baseline performances, was not optimized for detecting visualization performance differences. Our results offer actionable insights for conducting evaluation studies in decision-centric visualization, informing both the development of metrics and the interpretation of comparative studies. A preprint of this paper, along with the pre-registration, data analysis scripts, and all supplemental materials, will be made available at https://osf.io/u63zw/?view_only=b8b04a7e505c45c89ae957674248c494.

**Index Terms**—Decision-making, multivariate visualization, preference elicitation, parallel coordinates, tabular visualization, user study.

## 1 List of Supplemental Materials

The preregistration for this study is available at `https://osf.io/kw8uj`.

The following supplemental materials are available on Github at `https://github.com/lcibulski/decision-study-supplementals/`:

- Questionnaire (cf. Section IV-D in the main paper): `Questionnaire.pdf`
- Screen captures of interactive stimuli in questionnaire: `interactive-stimuli/`
- Data: `data/participants-raw.csv` and `data/participants-preprocessed.csv` (accuracy scores for chosen items, attention score, training scores, and Mini-VLAT score extracted from raw data)
- Data analysis scripts: `data-analysis/`

The following supplemental materials are available in this document:

- Planned analyses (cf. Section V-A in the main paper)
  - H5 ANOVA (Section 3.1)
  - Results of choice assessment and technique preference (Section 3.2)
- Exploratory analyses (cf. Section V-B in the main paper)
  - Distributions for perceived cognitive load and perceived precision (Section 4.1)
  - Results of repeated analyses with filtered participant subset (Section 4.2)
  - Results of repeated analyses grouped by Mini-VLAT scores (Section 4.3)

All supplemental materials will be made available on OSF at `https://osf.io/u63zw/?view_only=b8b04a7e505c45c89ae957674248c494`.


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## 2 Experiment Procedure - Questionnaire

Our experiment procedure consisted of five phases: Mini-VLAT, training, preference elicitation, decision-making (including assessment), and post-task survey. An equal number of participants was invited for each of the following sequences (visualization / dataset / elicitation):

1. PC / dataset 1 / low expressiveness then TV / dataset 2 / low expressiveness
2. TV / dataset 1 / low expressiveness then PC / dataset 2 / low expressiveness
3. PC / dataset 1 / medium expressiveness then TV / dataset 2 / medium expressiveness
4. TV / dataset 1 / medium expressiveness then PC / dataset 2 / medium expressiveness
5. PC / dataset 1 / high expressiveness then TV / dataset 2 / high expressiveness
6. TV / dataset 1 / high expressiveness then PC / dataset 2 / high expressiveness

A document with the questionnaire for all three expressiveness levels (i.e., sequences 2, 4, and 6) can be found at `supplementals-archive/Questionnaire.pdf`. Where participants are directed along different routes depending on their assigned expressiveness level, we concatenate the pages belonging to each of the routes at the respective location in the questionnaire. We explicitly label those pages that are only shown to a particular expressiveness condition.

To convey the interaction with the function editors at the medium and high expressiveness levels as well as with the parallel coordinates and tabular visualizations, we provide additional screen captures at `supplementals-archive/interactive-stimuli/`.

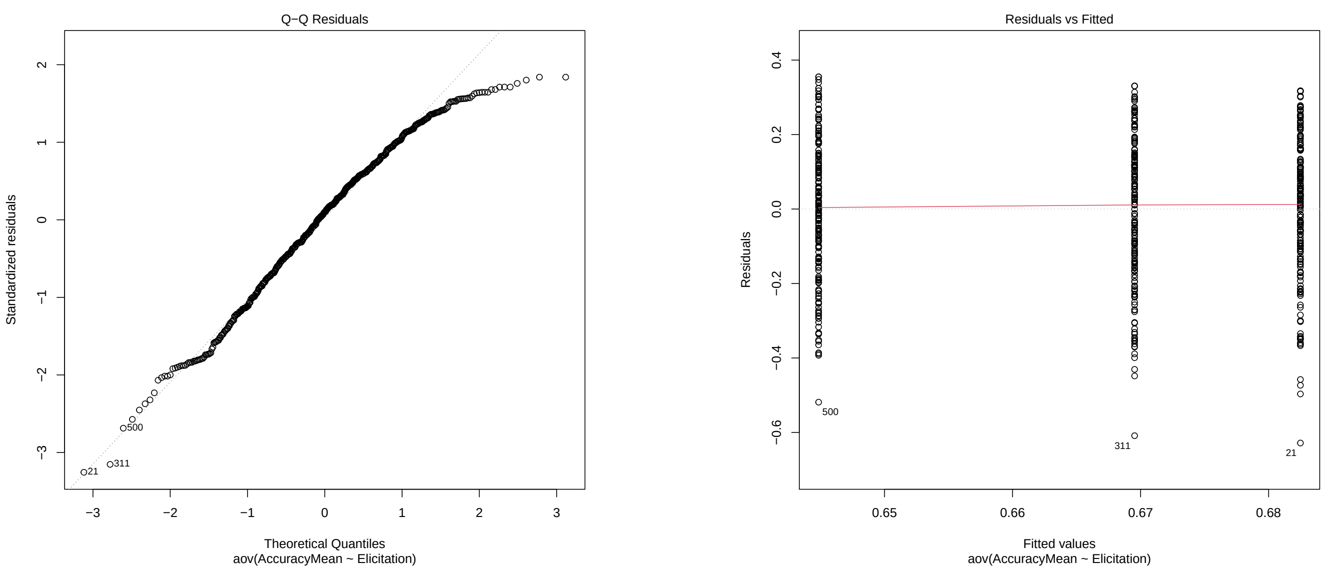
## 3 Planned Confirmatory Analyses

### 3.1 H5 ANOVA

To test hypothesis 5, i.e., whether using a preference elicitation interface with medium or high expressiveness results in less noise in the downstream metric, we observe how the variability of decision accuracy scores changes in response to varying expressiveness levels. We conducted a one-way ANOVA test to complement the comparison of boxplots in the main paper (Figure 1).

```
             Df Sum Sq Mean Sq F value Pr(>F)
Elicitation   2  0.133 0.06665   1.779   0.17
Residuals   545 20.424 0.03748
```

(a) ANOVA model



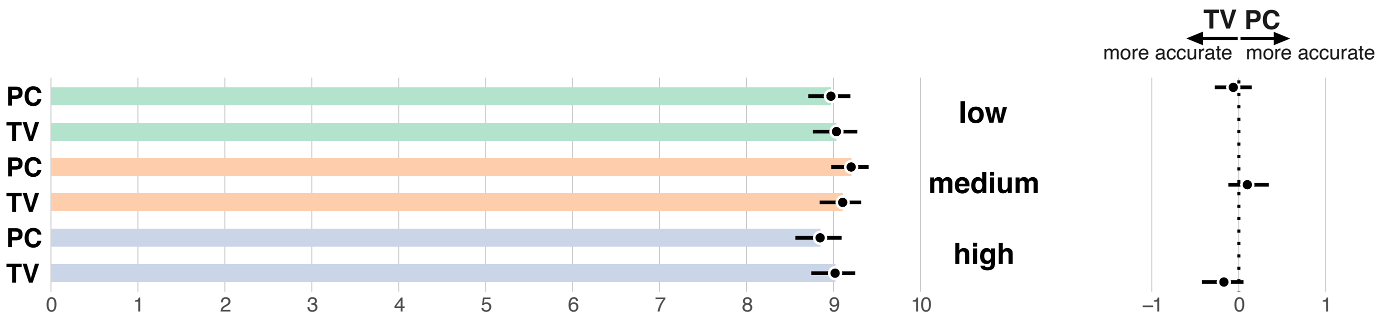

(b) Violated Normality Assumption

(c) Violated Equal Variances Assumption

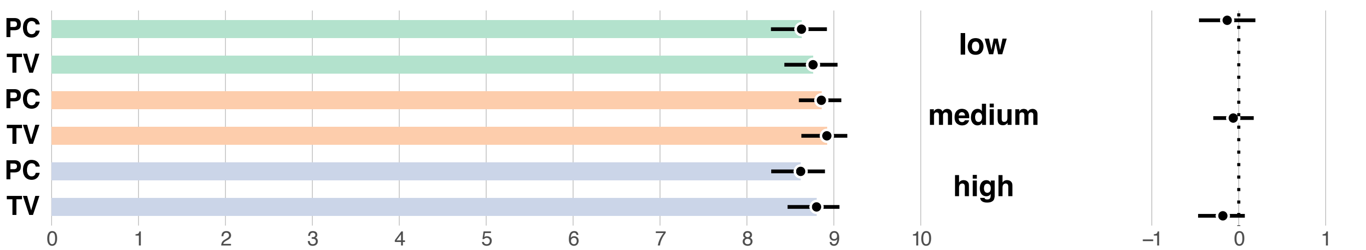
Fig. 1: One-way ANOVA to test $H_5$, i.e., whether variability in responses is lower when using preference elicitation with medium or high expressiveness as compared to low expressiveness. A p-value of 0.17 does not hint at a significant effect (a). However, both the normality (b) and equal variances (c) assumptions are violated. The independence assumption is met by randomly assigning participants to elicitation groups.

### 3.2 Subjective Choice Assessment and Technique Preference

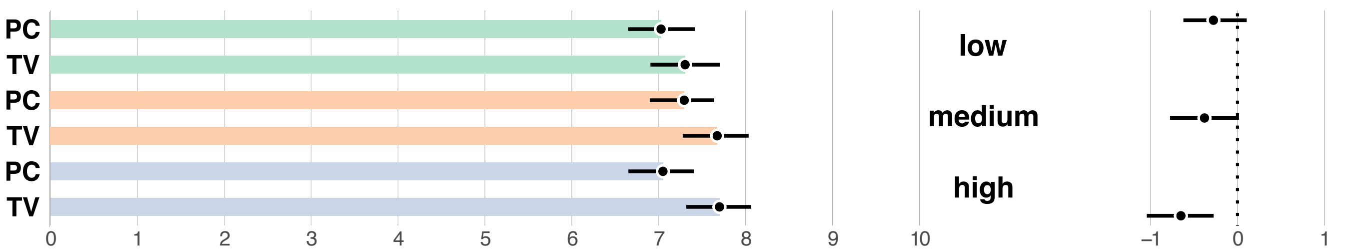
After each decision task, participants rated four different choice assessment metrics. After completion of both decision tasks, they additionally rated how helpful they found each technique to make their decisions. Here, we provide interval estimates of mean ratings and mean differences for each choice assessment metric and visualization preference (Figure 2).



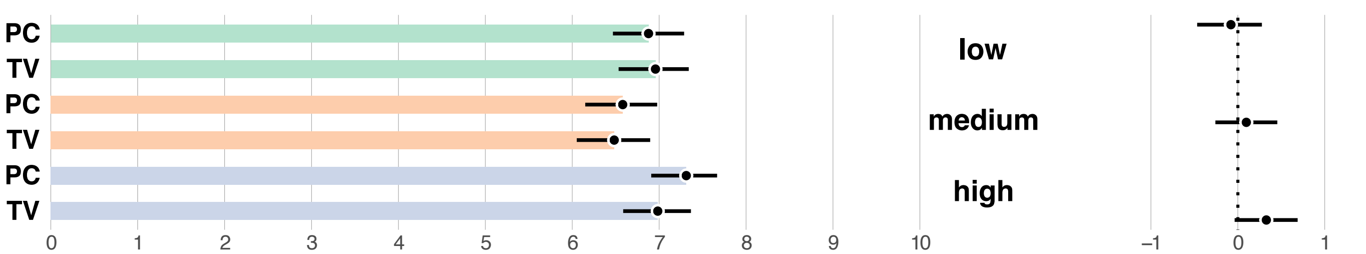

(a) Satisfaction



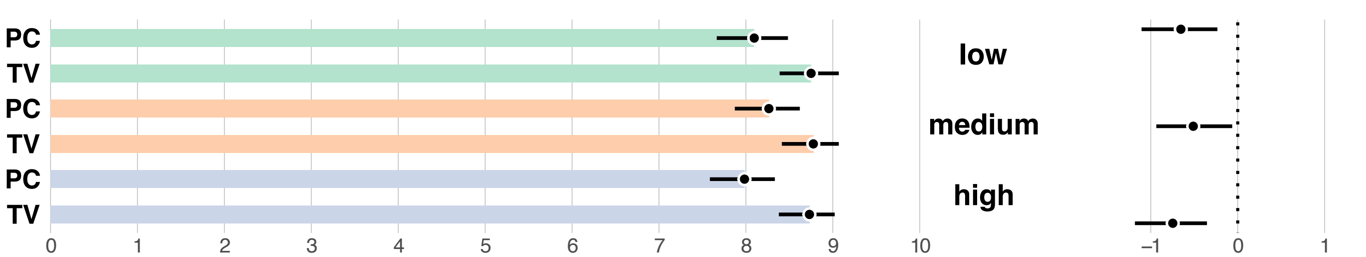

(b) Confidence




(c) Easiness




(d) Attachment



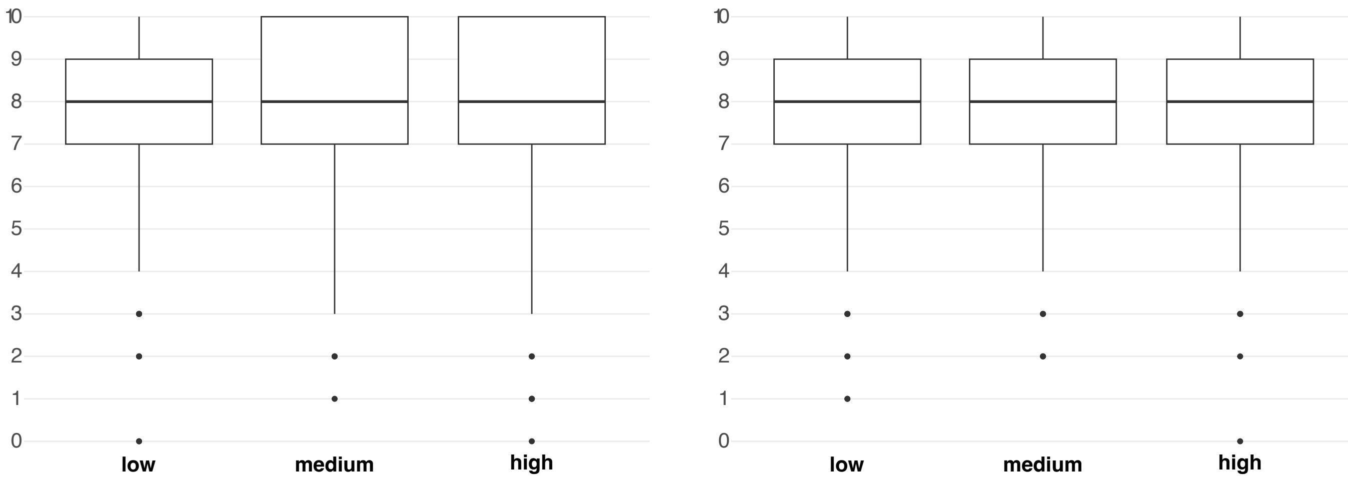

(e) Technique Preference

Fig. 2: Mean rating for each choice assessment metric and technique preference across each visualization and expressiveness condition (left). Mean differences between each pair of visualizations per expressiveness condition (right). All error bars are 95% CIs (n=188 for low, n=174 for medium, and n=186 for high expressiveness).

# 4 Additional Exploratory Analyses

## 4.1 Perceived Cognitive Load and Precision

To gain an understanding whether and how different cognitive loads or perceived precision associated with the expressiveness levels might have influenced our observed results, we analyzed the distributions of self-reported cognitive load and perceived precision across all three expressiveness levels (Figure 3).




(a) Perceived Cognitive Load (b) Perceived Precision

Fig. 3: Boxplots summarizing the variation of responses for perceived cognitive load (a) and perceived precision (b) across low, medium, and high expressiveness levels for preference elicitation.

## 4.2 Repeated Analyses with Filtered Participant Subset

We wanted to know whether our results involved statistical noise that originates from participants not knowing how to work with the visualizations or preference elicitation interfaces. We therefore repeated the planned analyses with only those participants who passed both attention checks, the preference elicitation training task (where applicable), and at least two of three training tasks for each visualization. This resulted in 305 participants (n=148 for low, n=78 for medium, and n=79 for high preference elicitation expressiveness), aged from younger than 20 to older than 60 with 72% aged between 20 and 39 years. 124 identified as female (40.7%) and 181 identified as male (59,3%). Results are shown in Figure 4.

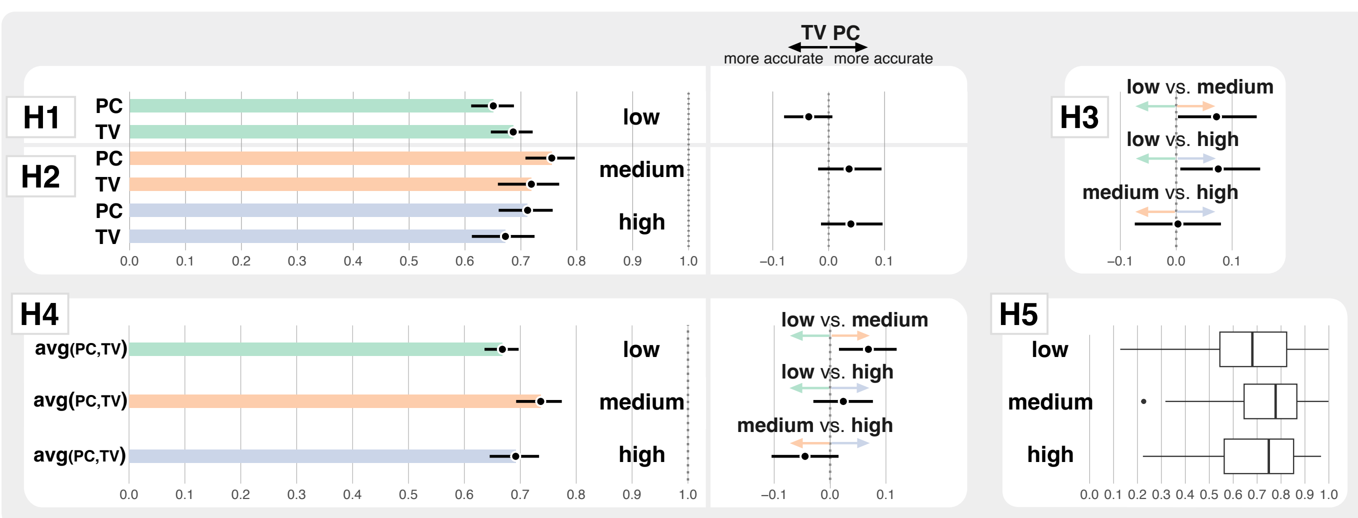



Fig. 4: The same 95% CIs as in Figure 4 of the main paper but only for those 305 participants (low n=148, medium n=78, high=79) who passed the attention checks and training tasks. *Left*: CIs with colored bars show mean decision accuracy per condition (H1, H2, H4). *Center*: Black CIs depict mean differences in accuracy between visualizations (H1, H2) and mean differences in overall accuracy between expressiveness levels (H4). *Right*: Black CIs depict the difference in PC-TV differences between all three expressiveness levels to analyze metric sensitivity (H3). Boxplots summarize the variation of measured accuracy scores for each expressiveness level (H5).

### 4.3 Repeated Analyses Grouped by Mini-VLAT Scores

To additionally control for visualization literacy as a potential explanation for statistical noise, we also repeated the analysis with all 548 participants grouped by their Mini-VLAT scores. Based on the distribution of scores (Figure 5), we grouped scores into low (0 to 4 with n=138), medium (5 to 7 with n=277), and high (8 to 12 with n=133). The results of the repeated analysis are shown in Figure 6.

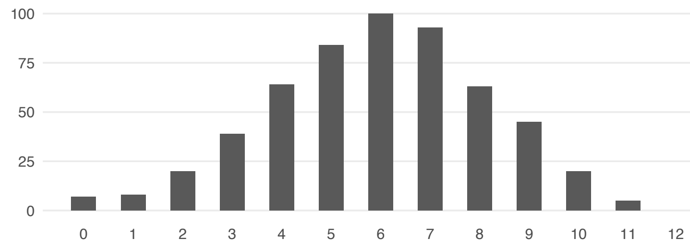



Fig. 5: Histogram of Mini-VLAT scores. Low scores are in [0,4] (n=138), medium scores in [5,7] (n=277), and high scores in [8,11] (n=133). In the low expressiveness condition, scores are distributed as follows: low n=46, medium n=103, high n=39. In the medium expressiveness condition, the distribution is: low n=46, medium n=74, high n=54. In the high expressiveness condition, the scores distribute as follows: low n=46, medium n=100, high n=40.

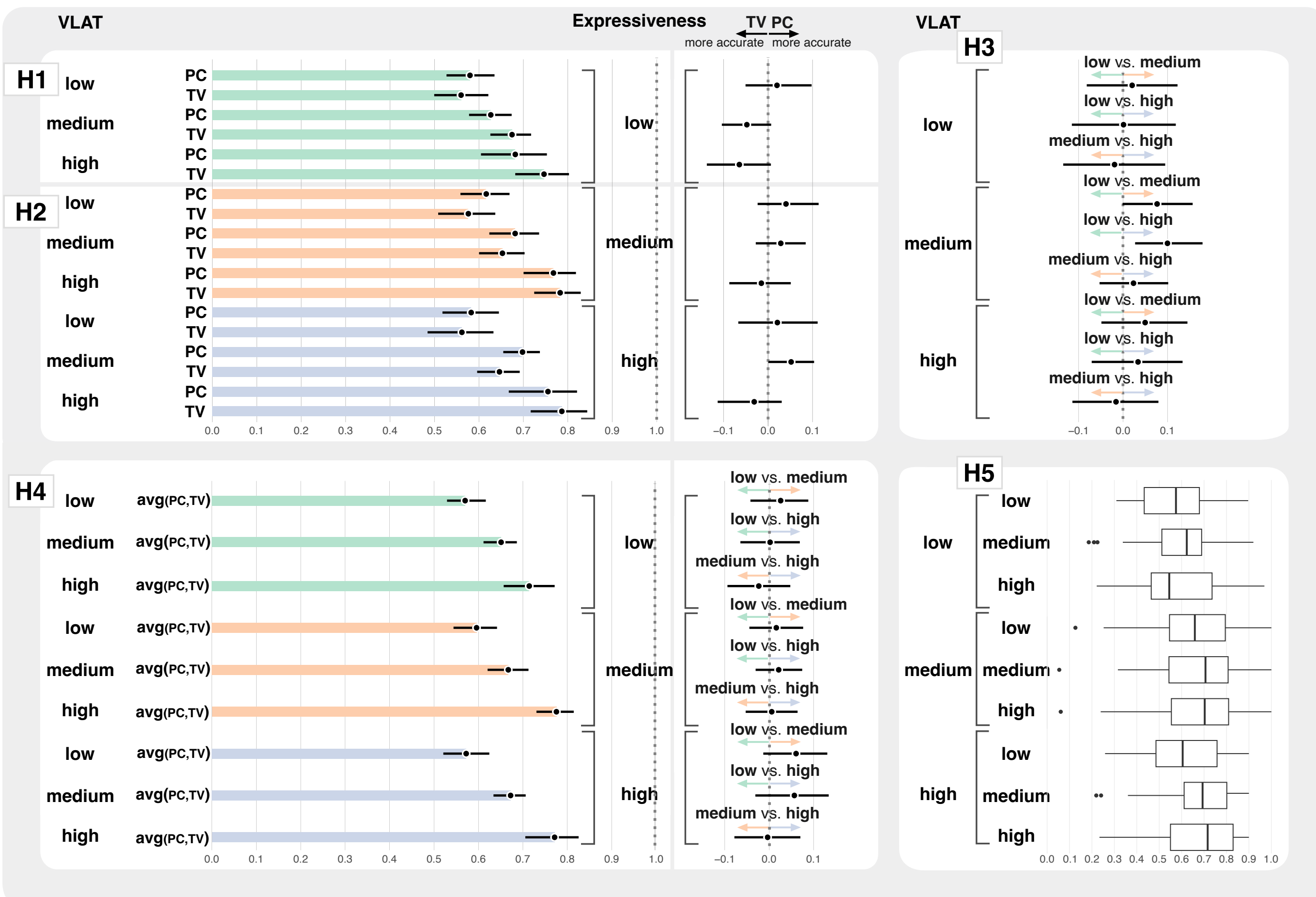



Fig. 6: The same 95% CIs as in Figure 4 of the main paper but separated into groups of low (n=138), medium (n=277), and high (n=133) Mini-VLAT scores. *Left*: CIs with colored bars show mean decision accuracy per condition (H1, H2, H4). *Center*: Black CIs depict mean differences in accuracy between visualizations (H1, H2) and mean differences in overall accuracy between expressiveness levels (H4). *Right*: For each Mini-VLAT level, black CIs depict the difference in PC-TV differences between all three expressiveness levels to analyze metric sensitivity (H3). For each Mini- VLAT level, boxplots summarize the variation of measured accuracy scores for each expressiveness level (H5).